

precog-data-intake: Tolls for automating bulk discovery, file checking, validation and download of Earth System Model archives across Earth System Federation nodes

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Summary

precog-data-intake is a Python command-line-interface (CLI) software for automated discovery, checking, validation, and download of Earth System Model outputs from Earth System Grid Federation (ESGF) archives. The software is designed for research workflows that require reproducible access to large, distributed climate-model datasets and is particularly aimed at bulk screening of Earth system archives before downstream analysis. The software builds on inherited ESGF access functionality from `intake-esgf` (Collier et al., 2026), while introducing interactive CLI workflows for archive interrogation, file-availability checks, grid-consistency cross-checks, temporal validation, export of search-result in tabular formats, and download management of shortlisted Earth System Model data products.

Statement of need

Modern Earth system science workflows frequently rely on climate-model data distributed across ESGF nodes. Although ESGF provides a federated infrastructure for searching and accessing these archives, practical research workflows often require additional automation to determine whether a given combination of variables, experiments, ensemble members, and grid configurations is both scientifically suitable and operationally downloadable before substantial time is spent retrieving files.

This issue is especially important for CMIP6-style analyses in which researchers may need to confirm that the same model provides paired pre-industrial, historical and ssp-scenario simulations, that multiple variables of interest are available on compatible grids for downstream analyses, and that temporal coverage is continuous across archived files. Manual inspection of catalogue search results can become slow, repetitive, and error-prone when screening many candidate models or variables across multiple ESGF nodes.

`intake-esgf` provides an important upstream foundation for this work by enabling access to information on ESGF nodes (Collier et al., 2026). `precog-data-intake` builds on that core implementation but targets a different level of the workflow: rather than focusing only on data access primitives and caching to in-memory use, it provides an interactive command-line environment for ample discovery, screening, validating and managing downloads of published ESGF archive products that satisfy scientific constraints relevant to Earth system and ocean biogeochemistry applications. The software is intended for situations in which researchers need to move efficiently from broad catalogue searches to a smaller, analysis-ready subset of model output that satisfies practical and scientific constraints.

Therefore, precog-data-intake fills a workflow gap between catalogue access and scientific analysis. It helps users shortlist Earth System Models that satisfy compound criteria across experiments and variables, validate archive consistency before download, export tabular summaries of search results, identify datasets that are actually downloadable, and organize retrieved files into a local directory structure suited to reproducible downstream analysis. This design is particularly useful when users need to screen many candidate models and variables before deciding which datasets are scientifically suitable and operationally downloadable.

Key functionality

The software provides several features tailored to archive-scale Earth system data workflows:

- automated ESGF searches using project, activity, experiment, frequency, variable, and grid filters (inherited from intake-esgf);
- interactive user prompts for download paths and variable selections within CLI workflows;
- bulk screening of CMIP6 archive holdings for paired piControl and historical availability;
- validation of continuity in date stamps for CMIP6 pre-industrial and historical simulations;
- checks for consistent availability of model output across compatible grids such as gn and gr;
- conditional multi-variable screening for models that satisfy compound search criteria, including cases where several required variables must be available simultaneously;
- export of tabular ESGF catalogue search results for easy inspection, logging, and reproducible shortlist generation;
- parallel URL pre-checks with verification of server responses and flagging of shortlisted outputs as downloadable;
- download manager with parallel transfers that select responsive ESGF node locations and include retry or integrity checks for corrupted files
- retrieval of ocean grid-cell measure variables such as areacello and volcello for selected models, together with saved archive snapshots for later inspection.

These capabilities support workflows in which data access is itself a significant part of the scientific process, particularly when analyses depend on assembling coherent, scientifically consistent, and analysis-ready ensemble subsets of Earth system model output from distributed archives.

Research applications

A representative use case is the identification of CMIP6 models that simultaneously provide both pre-industrial and historical output for ocean biogeochemical variables such as expc and epc100, together with auxiliary or supporting variables and the associated grid-cell measures required for downstream calculations. The repository accompanying the software includes a [workflow example](#) demonstrating this type of archive screening and retrieval process.

This is particularly relevant for ocean biogeochemistry and carbon-cycle studies, where analyses often depend on coherent combinations of physical and biogeochemical fields rather than isolated variables (e.g., retrieval of carbonate system fields as well as ocean state physical variables). In such cases, the time spent screening archive holdings, checking consistency, and organizing downloads can be substantial, and purpose-built automation improves both efficiency and reproducibility.

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90 References

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